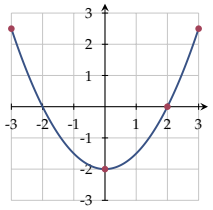
Sheet 72 Tue 25 May 2027 *From table to tree and symbolic notation*

1. Warm-up · Quick Express $\dfrac{30}{60}$ as a probability.
2. Core · Method From the table on sheet 69, build the tree (F or G, then S or $\overline{S}$): give $P(F)$, $P_F(S)$ and $P_G(S)$.
3. Core + · Translate Write symbolically and calculate "the probability of being a boy who does not do sport".
4. Review · Quick A car travels at constant speed: 150 km in 2 h. What distance in 5 h?
5. Thread · Statement → formula A taxi charges a pick-up fee of 2 000 Ar then 1 500 Ar per km. Express the price $p(x)$ for x km, then calculate $p(8)$.

Sheet 73 Thu 27 May 2027 *Reversing conditioning: $P_A(B)$ and $P_B(A)$*

1. Warm-up · Quick What is the difference between $P_M(T)$ and $P_T(M)$?
2. Core · Method Using the data from sheet 70 ($P(M \cap T) = 0.27$ and $P(T) = 0.305$), calculate $P_T(M)$.
3. Core + · Method Rare disease (1 % of the population); the test is positive for 99 % of those with the disease and 5 % of those without. Calculate $P_T(M)$.
4. Review · Graph Curve of f:



   Read $f(2)$ and the preimages of 2.5.
5. Thread · Method Program: choose a number; multiply it by -2; add 10; divide by 2. Result for 7? Expression in x?

Sheet 74 Mon 31 May 2027 *False positives and false negatives*

1. Warm-up · Quick Define "false positive" and "false negative".
2. Core · Method 10 000 people, 2 % with the disease; the test is positive for 90 % of those with the disease and 10 % of those without. How many false negatives, false positives and positives in total?
3. Core + · Interpret Using the previous data, calculate the proportion of people who truly have the disease among the positives and comment.
4. Review · True/false True or false? Justify. The area of a square is proportional to the length of its side.
5. Thread · Disprove Conjecture: "for every integer n, $n^2 + n + 41$ is prime". Test $n = 40$.

Consolidation, starter routines and transition to Year 12 (Première)

Sheet 75 Tue 1st June 2027 *Synthesis problems drawing on several topics*

1. Warm-up · Quick Solve $3(x - 1) = x + 5$.
2. Core · Method For which value of k are the points $A(1;1)$, $B(3;5)$ and $C(k;9)$ collinear?
3. Core + · Method A price increases by 10 % then decreases by 10 %. What is the overall change?
4. Review · Quick Total surface area of a cuboid with dimensions 4, 3, 2 (cm)?
5. Thread · Quick Calculate the exact volume of a sphere of radius 3 cm.

Sheet 76 Thu 3 June 2027 *Review of the six competences*

1. Warm-up · Metacognition List the six mathematical competences developed at lycée.
2. Core · Metacognition Match each of the following to a competence: "translating a problem statement into an equation"; "sketching a curve"; "justifying an equality with a written calculation".
3. Core + · Metacognition To solve $2x + 3 = 11$, which competence is mainly used?
4. Review · Problem Water at 20 °C is heated by 3 °C per minute: $T(t) = 20 + 3t$. After how many minutes does it reach 65 °C?
5. Thread · Quick Solve in $\mathbb{R}$: $0.5x + 1.2 = 2.7$.

Sheet 77 Mon 7 June 2027 *Targeted reactivation of prerequisites for Première*

1. Warm-up · Quick Expand $(x + 3)^2$.
2. Core · Method Check that $x^2 - 5x + 6 = (x - 2)(x - 3)$ and deduce the solutions of $x^2 - 5x + 6 = 0$.
3. Core + · Anticipation For $f(x) = x^2$, calculate the rate of change $\dfrac{f(3) - f(1)}{3 - 1}$.
4. Review · Construction Place M and N, the midpoints of $[AB]$ and $[AC]$, then draw $[MN]$. What can be said about (MN) and (BC)? Calculate MN if $BC = 9$ cm.
5. Thread · Write-up Prove that a triangle with sides 7, 24 and 25 is right-angled.

Sheet 78 Tue 8 June 2027 *Synthesis problems: choosing and justifying a strategy*

1. Warm-up · Reading code What does $f(2)$ return?

```
def f(x):
    return x**2 - 3*x + 1
```

2. Core · Completion Complete the code so that the function returns `True` when A, B, C are collinear:

```
def alignes(xa, ya, xb, yb, xc, yc):
    return (xb-xa)*(yc-ya) - (yb-ya)*(xc-xa) == ....
```

3. Core + · Completion Complete the absolute value function:

```
def vabs(x):
    if x >= 0:
        return x
    else:
        return ....
```

4. Review · Quick Write 2.5×10^3 and then 6.2×10^{-2} in decimal notation.
5. Thread · Quick Give the reflection of $A(2; -3)$ in the origin.